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Ceruloplasmin deficiency drives a fusiform-centric lipid–myelin pathology underlying a visual subtype in autism

Ya-Yin Deng[1,5], Shuang-Shuang Zhong[1,5], Shi-Huan Wang[2,5], Bo-Ya Yin[1,5], Xiang Zhou[1], Feng-Yun Zou[1], Jia-Yuan Zhao[1], Yu-Xuan Ni[1], Xiao-Wen Luo[1], Li-Shan Shen[1], Jia-Lu Zhang[3], Zhi-Ping Lin[3], Wen-Ying Zhou[4]✉, Hong-Zhu Deng[2]✉ and Ruo-Mi Guo[1]✉

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Atypical visual processing (AVP) commonly occurs in autism spectrum disorder (ASD) and contributes to social impairments, yet its neurobiological basis remains poorly characterized. To investigate potential lipid and myelin mechanisms, this study integrated multimodal MRI, quantifying lipid (proton density fat fraction) and myelin content (synthetic MRI) of 74 nuclei or brain regions, with serum profiling of iron, lead, and ceruloplasmin in 288 children, including 90 ASD with atypical visual processing (ASD-AVP), 89 ASD without atypical visual processing (ASD-AVP), and 109 typically developing (TD). The ASD-AVP subgroup exhibited a distinct co-pathology of elevated lipid and myelin centered on the fusiform gyrus (FG), accompanied by a unique positive lipid-myelin correlation (left: $r = 0.47$, right: $r = 0.41$). Serum analyses revealed decreased iron and ceruloplasmin and increased lead in ASD-AVP, and mediation analysis indicated that ceruloplasmin deficiency influences FG myelination via lipid pathways (35%–55%). Crucially, BTBR AVP-like mice recapitulated this phenotype with disorganized hypermyelination, whereas nAVP-like mice showed hypomyelination. A combined FG lipid-myelin signature strongly distinguished ASD-AVP from TD (AUC = 0.93) and ASD-nAVP (AUC = 0.87). Preliminary longitudinal follow-up in a subset of patients revealed that improvement in serum ceruloplasmin was associated with a reduction in FG lipid content and stabilization of myelin, paralleling clinical improvement. These findings identify a ceruloplasmin-driven, FG-centric lipid-myelin co-pathology, representing a maladaptive "inflammatory pseudo-compensation" mechanism specific to a visual ASD subtype, and offer novel biomarkers for biological subtyping and targeted interventions.

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INTRODUCTION

Autism spectrum disorder (ASD) is a complex neurodevelopmental condition characterized by differences in social communication, language impairments, restricted interests, and repetitive stereotyped behaviors [1]. Its rising prevalence has created substantial social, economic, and healthcare burdens worldwide [2]. Sensory abnormalities are common in ASD [3], yet often underrecognized. Among these, visual processing anomalies are particularly relevant to core social deficits in ASD [4].

Individuals with ASD often display atypical visual behaviors, including light hypersensitivity, local processing bias, and avoidance of eye contact [5]. These impairments reduce global information integration and hinder social functioning by disrupting facial expression decoding and social intention understanding [6]. Despite its early onset and clinical significance, investigations on atypical visual processing have focused mainly on ocular anatomy, localized functional MRI (fMRI), and structural MRI (sMRI) [5–7]. The absence of systematic whole-brain multi-scale and multi-parametric analyses has limited deeper pathophysiological understanding of visual processing anomalies in ASD.

Lipids, composing about 50% of the human brain's dry weight, are essential structural components of neuronal membranes. Disrupted lipid homeostasis is implicated in various neurological disorders [8] and may provide both mechanistic insights and diagnostic biomarkers for ASD [9, 10]. The myelin sheath, with the highest lipid content in the central nervous system (70%–80% dry weight) [11], is frequently altered in ASD, and dysmyelination is linked to its pathophysiology [12–14]. Lipid imbalances influence myelination bidirectionally: deficiency impedes myelin formation, while excess promotes oxidative stress and myelin damage [15]. Elevated cerebral oxidative stress is a potential trigger for ASD [16]. Mapping interactions between lipid metabolism, myelination, and oxidative stress in specific brain regions could therefore yield novel insights into ASD pathogenesis and early diagnosis.

MRI has advanced biomarker discovery for ASD [17]. While magnetic resonance spectroscopy (MRS) remains the gold standard for lipid quantification, it is limited in spatial coverage. Dixon-based proton density fat fraction (PDFF) mapping offers superior detection of subtle lipid abnormalities across brain regions [18]. Additionally, synthetic MRI (SyMRI) enables sensitive and direct myelin quantification in children [12]. Integrating these

[1]Department of Radiology, The Third Affiliated Hospital of Sun Yat-Sen University, No. 600 Tianhe Road, Guangzhou 510630, China. [2]Department of Child Development and Behavior Center, the Third Affiliated Hospital of Sun Yat-Sen University, No. 600 Tianhe Road, Guangzhou 510630, China. [3]Research Department, GE Healthcare, Beijing, People's Republic of China. [4]Department of Clinical Laboratory, The Third Affiliated Hospital of Sun Yat-sen University, No. 600 Tianhe Road, Guangzhou 510630, China. [5]These authors contributed equally: Ya-Yin Deng, Shuang-Shuang Zhong, Shi-Huan Wang, Bo-Ya Yin. ✉email: zhouwy3@mail.sysu.edu.cn; denghzh@mail.sysu.edu.cn; guoruomi86@mail.sysu.edu.cn

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imaging techniques allows a comprehensive, region-specific assessment of lipid metabolism and myelin integrity.

To link metabolic dysregulation and neural alterations in ASD, peripheral biomarkers such as serum iron, ceruloplasmin, and lead levels provide valuable information. For instance, iron deficiency impairs myelination [19], and reductions in both iron and myelin correlate negatively with ASD symptom severity [20]. Ceruloplasmin, essential for iron metabolism, facilitates iron oxidation and its dysfunction may cause iron dysregulation and lipid peroxidation [21, 22]. Besides, neurotoxic lead can exacerbate cerebral iron deficiency and amplify oxidative stress [23, 24]. Together, these markers may provide a peripheral perspective on central visual pathway vulnerability in ASD.

Here, we pioneered a multimodal cross-scale framework that integrates PDFF-measured fat content, syMRI-derived myelin quantification, and serum iron-ceruloplasmin-lead profiling for ASD investigation, revealing a metal metabolism-lipid-myelin coupling pathology specific to the visual subtypes of ASD. Critically, we closed the loop between clinical imaging and pathological mechanism by demonstrating parallel, histologically validated myelin alterations in a BTBR mouse model, thereby strengthening the biological plausibility of our findings.

METHODS AND MATERIALS

Participants

Between May 2022 and June 2025, we prospectively enrolled 202 children with ASD (aged 6–12 years) and 116 age- and sex-matched typically developing (TD) controls. This study was approved by the Institutional Review Board of The Third Affiliated Hospital of Sun Yat-sen University (Approval Number: [2022]02-005-01; February 2022). Written informed consent was obtained from parents or legal guardians.

Demographic and clinical data were collected for all participants, including age, sex, handedness, and intelligence quotient (IQ). ASD diagnoses were confirmed by two senior clinicians according to DSM-5 criteria [25], supported by the Autism Diagnostic Observation Schedule–Calibrated Severity Score (ADOS-CSS) and Child Sensory Profile 2 (SP2) [26]. Based on SP2 visual scores, ASD participants were divided into two groups: (1) ASD-AVP (atypical visual processing), defined as SP2 raw score ≥ TD mean + 1 SD (≥ 16.80), indicating visual hypersensitivity [27]; (2) ASD-nAVP (without atypical visual processing), with SP2 scores within ± 1 SD of the TD mean (9.80–16.79). Participants scoring < 9.80 were excluded to avoid heterogeneity from visual hyposensitivity. TD reference SP2 values were derived from all controls after exclusion (mean visual score = 13.30 ± 3.50).

Exclusion criteria for all participants (ASD and TD) groups included: (1) unstable or chronic medical conditions; (2) metal implants, neurological or psychiatric disorders, structural abnormalities, or known genetic syndromes; and (3) use of psychotropic medications. Additionally, participants were excluded if they were taking any non-psychotropic medications or dietary supplements known to significantly impact neurometabolism, metal levels, or lipid profiles (e.g., iron supplements, omega-3 fatty acids, systemic corticosteroids, or anticonvulsants for non-psychiatric indications). All medication and supplement use was thoroughly reviewed via parent interview and medical record check. TD participants had no personal or familial history (including in first-degree relatives) of mental illness, neurodevelopmental disorders, craniocerebral disease, chronic conditions, or substance abuse, as assessed through clinical interviews. All included participants were right-handed; left-handedness was not an exclusion criterion, but no left-handed individuals were present in the final sample. All MRI scans were reviewed and confirmed normal by neuroradiologists. Participants with poor image quality or significant motion artifacts were excluded.

To preliminarily assess the dynamic relationship between serum ceruloplasmin, clinical symptoms, and brain measures, we conducted a 2-year follow-up in a convenience subset of seven children from the ASD-AVP group (all males; mean baseline age 7.71 ± 1.60 years). Follow-up evaluations included repeated serum ceruloplasmin measurement, SP2 assessment, and identical multimodal MRI protocols for PDFF and MyC quantification.

MRI data acquisition

All MRI data were acquired on a 3.0 T scanner (SIGNA™ Architect, GE Healthcare, USA) with a 48-channel head coil. MRI scans were conducted during natural sleep to minimize motion. The scan protocol included: high-resolution 3D T1-weighted imaging, a 3D multi-echo Dixon sequence (IDEAL-IQ) for proton density fat fraction (PDFF) mapping, and synthetic MRI (SyMRI) using a 2D multi-dynamic multi-echo sequence with two echoes and four delay times per slice. Detailed acquisition parameters are listed in Table S1. The reliability of the PDFF technique for lipid quantification was validated in vitro in a prior study [28].

MRI data processing

PDFF maps were generated automatically from IDEAL-IQ acquisitions. SyMRI data were processed using SyntheticMR software (v11.2.2, SyntheticMR AB, Sweden) to produce quantitative maps of R1 (1/T1), R2 (1/T2), and proton density (PD). Tissue segmentation included white matter (WM), gray matter (GM), cerebrospinal fluid (CSF), and non-tissue (NoN) [29]. Voxel-wise myelin volume fraction maps were computed using a four-compartment model including myelin, cellular content, free water, and excess parenchymal water [30].

PDFF maps, water-only IDEAL-IQ images, SyMRI MyC maps, and 3D T1-weighted images were exported for further analysis. Data processing was performed using an in-house Python pipeline. Both PDFF and MyC maps were co-registered to the 3D T1-weighted image using the water-only image and the synthetic T1-weighted image as reference, respectively. High-resolution 3D T1-weighted images were segmented into 74 brain regions using FastSurfer according to the Desikan-Killiany atlas [31]. Mean values of PDFF and MyC from all segmented regions were extracted for further statistical analyses [31].

Blood biomarker quantification

Fasting blood samples were collected from all participants using trace-metal-optimized procedures. After an 8-hour overnight fast, venipuncture was performed between 8:00–10:00 AM using IMPROVACUTER® collection tubes (EDTA.K2, no additive, and heparin tubes). Skin was cleansed with alcohol and metal-free saline. Tubes were gently inverted and centrifuged at 1500 rpm for 10 min within 30 min of collection. Serum and whole blood were aliquoted into pre-screened containers; ceruloplasmin samples were stored in Eppendorf Safe-Lock™ tubes. Hemolyzed samples ($n = 2$) were excluded. All aliquots were stored at −20°C with ≤ 2 freeze-thaw cycles. Behavioral support was provided to children with ASD to reduce distress. Assay methods for iron, ceruloplasmin, and lead are provided in Table S2. As a result, serum iron, ceruloplasmin, and lead levels were obtained for all participants.

Statistical analysis

Data normality was assessed using the Shapiro–Wilk test in GraphPad Prism (v9.5.1). Normally distributed data were presented as mean ± SD, and non-normal data as median and interquartile range (IQR). Sex distribution differences were evaluated with the Chi-square test.

Between-group comparisons of MRI metrics. One-way ANOVA was performed for each of the 74 brain regions to assess between-group differences of MRI-derived metrics, PDFF, and MyC, using Python 3.10 (SciPy 1.9.0). Multiple comparisons were corrected using the Benjamini–Hochberg false discovery rate (FDR). For regions with significant between-group differences, post-hoc Tukey's HSD tests were applied, controlling the family-wise error rate (FWER) at $\alpha = 0.05$ per region.

Correlation analyses of MRI metrics. Pearson or Spearman (based on data distribution) were computed between PDFF and MyC values within regions exhibiting significant between-group differences. Analyses were performed in GraphPad Prism (v9.5.1), with age and sex as covariates. Benjamini–Hochberg FDR was used for multiple comparison correction. Correlations with $|r| \geq 0.3$ and FDR-corrected $p < 0.05$ were considered biologically significant.

Blood biomarker analyses. Inter-group differences in serum iron, ceruloplasmin, and lead were assessed after testing for normality (Shapiro–Wilk) and variance homogeneity (Brown–Forsythe) using GraphPad Prism (v9.5.1). Parametric data were analyzed with one-way ANOVA and Tukey's post hoc test; non-parametric data with Kruskal–Wallis and Dunn's tests. Spearman correlations were computed between each serum biomarker and MRI-derived PDFF/MyC values from all 74 regions with Benjamini–Hochberg FDR correction. Significant associations were defined as $|r| \geq 0.3$ and FDR-corrected $p < 0.05$. Pairwise correlations among biomarkers were also examined within each group.

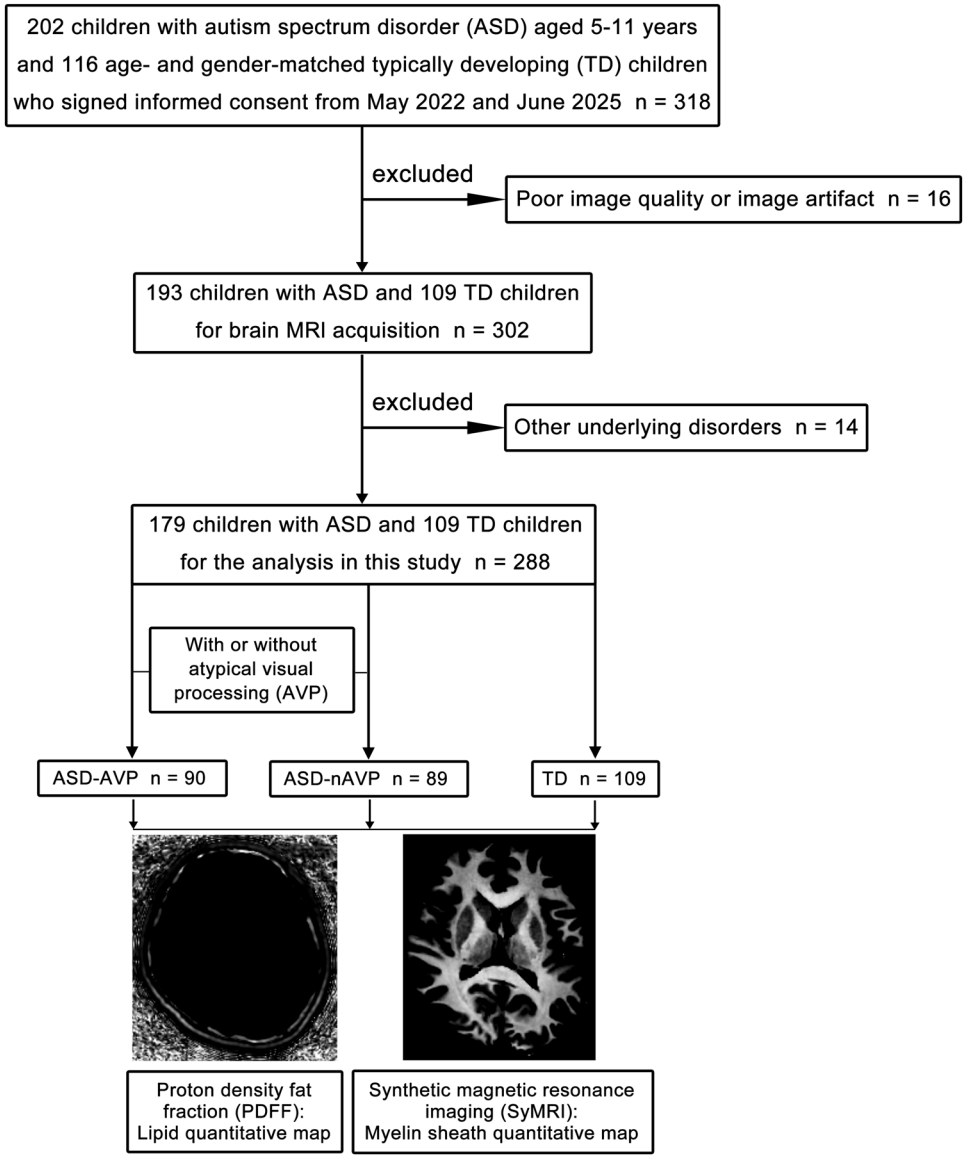

Fig. 1 Study flow and group allocation. Flowchart of participant inclusion and exclusion.

Mediation analysis. For the ASD-AVP group, causal mediation analysis was employed to assess the potential mediating role of regional lipid content in the relationship between blood biomarkers and myelin content. In the ASD-AVP group, mediation models tested whether regional lipid content mediated the effect of blood biomarkers on myelin content. Regions with significant ANOVA results in both lipid and myelin measures were included. For each region, three models were constructed using serum biomarker level as an independent variable (X), lipid content as a mediator (M), and myelin content as an outcome (Y). Analyses were conducted in R 4.3.1 (mediation package v4.3.0) with non-parametric bootstrapping (1000 samples, $p < 0.05$).

Diagnostic performance. To further validate the importance of lipid and myelin contents in pathophysiological mechanisms of ASD, receiver operating characteristic (ROC) analysis was performed for diagnostic performance assessment using lipid and myelin contents in specific regions as model regressors. Bonferroni correction was applied for multiple comparisons (corrected $\alpha = 0.017$).

In vivo validation of brain pathology in BTBR mice

To further validate our MRI findings, we employed a novel visual shape discrimination paradigm to stratify BTBR T(+) Itpr3(tf)/J mice (Research Resource Identifier: IMSR_JAX: 002282), a well-established autism model, based on dissociable visual-cognitive phenotypes. Myelin integrity in the fusiform gyrus (FG) was subsequently evaluated using Luxol Fast Blue (LFB) staining in both BTBR and control C57BL/6 J mice (see Supplementary Material). All animal procedures were approved by the Institutional Animal Care and Use Committee (IACUC) of The Third Affiliated Hospital of Sun Yat-Sen University (Approval No. IACUC-2023-A014) and were conducted in accordance with the relevant institutional guidelines and regulations and the National Institutes of Health Guide for the Care and Use of Laboratory Animals.

RESULTS

Participant characteristics

A total of 179 children with ASD (90 ASD-AVP and 89 ASD-nAVP) and 109 TD children were included (see Fig. 1 for exclusion details). The ASD-AVP group consisted of 66 males and 24 females (mean age 8.94 ± 1.52 years), the ASD-nAVP group included 67 males and 22 females (mean age 8.83 ± 1.49 years), and the TD group comprised 81 males and 28 females (mean age 9.02 ± 1.47 years). Demographic and clinical characteristics are summarized in Table 1. No significant differences were observed among the three groups in age, sex distribution, or BMI *z*-score (all $p > 0.05$).

Table 1. Clinical characteristics of participants in subgroups.

Characteristic	ASD-AVP (n = 90)	ASD-nAVP (*n* = 89)	TD (*n* = 109)	*p*-value (two-sample t-tests / ANOVA/χ^2)	Tukey-adjusted *p*-value		
					ASD-AVP vs. ASD-nAVP	ASD- AVP vs. TD	ASD-nAVP vs. TD
Gender (M/F)	66 M/24 F	67 M/22 F	81 M/28 F	0.89	/	/	/
Age (years)	8.94 ± 1.52	8.83 ± 1.49	9.02 ± 1.47	0.34	/	/	/
Handedness (right/left)	90/0	89/0	109/0	>0.99	/	/	/
BMI *z*-score	0.32 ± 1.10	0.29 ± 1.08	0.30 ± 1.05	0.94			
Iron level (umol L^{-1})	7.25 ± 2.63	21.33 ± 10.14	19.63 ± 3.36	**<0.01***	**<0.01***	**<0.01***	0.13
Ceruloplasmin level (g/L)	0.22 ± 0.05	0.27 ± 0.06	0.28 ± 0.05	**<0.01***	**<0.01***	**<0.01***	0.93
Blood lead levels (ug/L)	34.54 ± 23.30	29.95 ± 19.13	23.67 ± 15.18	**<0.01***	0.25	**<0.01***	0.06
ADOS-CSS score	8 (6−9)	7 (6−8)	/	>0.99	/	/	/
Visual section score of SP2	17.50 ± 3.60	13.10 ± 3.40	13.30 ± 3.50	**<0.01***	**<0.01***	**<0.01***	0.82

Values are presented as means ± standard deviations or median (interquartile range, IQR).
M/F male/female, χ^2 chi-square test, *ASD-AVP* autism spectrum disorder with atypical visual processing, *ASD-nAVP* autism spectrum disorder without atypical visual processing, *TD* typically developing, *ADOS-CSS* autism diagnostic observation schedule-calibrated severity score, SP2 child sensory profile 2.
Statistically significant *p*-values are in bold.
*$p < 0.05$ represents statistically significant.
*Tukey-adjusted *p*-value <0.05 represents statistically significant.

Table 2. PDFF differences among the three groups.

	ASD-AVP (n = 90)	ASD-nAVP (n = 89)	TD (n = 109)	one-way ANOVA (*Pfdr* /η^2)		*p*-value (Turkey HSD test)/Cohen's d					
						a		b		c	
PCAL_L	0.61 ± 0.12	0.57 ± 0.07	0.62 ± 0.11	**0.03***	0.04	**0.02***	0.43	>0.99	−0.02	**<0.01***	−0.48
PCAL_R	0.58 ± 0.11	0.54 ± 0.09	0.58 ± 0.12	**0.05***	0.03	**0.03***	0.42	0.99	0.02	**0.03***	−0.38
CUN_L	0.49 ± 0.12	0.44 ± 0.09	0.48 ± 0.13	**0.04***	0.03	**0.01***	0.46	0.84	0.07	**0.04***	−0.36
CUN_R	0.55 ± 0.14	0.49 ± 0.09	0.53 ± 0.10	**0.03***	0.04	**<0.01***	0.46	0.31	0.19	0.13	−0.32
LING_L	0.68 ± 0.16	0.61 ± 0.11	0.57 ± 0.11	**<0.01***	0.12	**<0.01***	0.52	**<0.01***	0.84	0.07	0.37
LING_R	0.61 ± 0.13	0.55 ± 0.12	0.51 ± 0.11	**<0.01***	0.09	**<0.01***	0.44	**<0.01***	0.81	**0.05***	0.35
FG_L	0.79 ± 0.15	0.55 ± 0.07	0.50 ± 0.13	**<0.01***	0.51	**<0.01***	2.06	**<0.01***	2.04	**0.04***	0.41
FG_R	0.75 ± 0.21	0.60 ± 0.16	0.53 ± 0.14	**<0.01***	0.22	**<0.01***	0.78	**<0.01***	1.23	**<0.01***	0.48
rACC_L	0.46 ± 0.17	0.39 ± 0.11	0.443 ± 0.128	**0.02***	0.04	**<0.01***	0.47	0.67	0.11	0.03	−0.42
rACC_R	0.44 ± 0.16	0.37 ± 0.12	0.426 ± 0.139	**0.02***	0.04	**<0.01***	0.50	0.81	0.08	**0.01***	−0.45
Pcu_L	0.49 ± 0.15	0.43 ± 0.08	0.46 ± 0.10	**0.02***	0.04	**<0.01***	0.50	0.26	0.20	0.08	−0.38
Pcu_R	0.51 ± 0.15	0.46 ± 0.07	0.49 ± 0.10	**0.05***	0.03	**0.01***	0.41	0.60	0.12	0.09	−0.38
rMFG_R	0.64 ± 0.65	0.48 ± 0.20	0.48 ± 0.25	**0.04***	0.03	**0.02***	0.33	**0.02***	0.33	>0.99	−0.02
POr_L	0.63 ± 0.62	0.45 ± 0.25	0.43 ± 0.33	**0.02***	0.04	**0.01***	0.39	**<0.01***	0.42	0.94	0.07
PaG_R	0.45 ± 0.12	0.43 ± 0.09	0.49 ± 0.15	**0.02***	0.04	0.5	0.18	0.05	−0.31	**<0.01***	−0.48
lOFC_R	1.30 ± 0.86	0.98 ± 0.59	1.01 ± 0.63	**0.02***	0.04	**<0.01***	0.43	**0.01***	0.38	0.93	−0.06
mOFC_L	1.02 ± 0.52	0.89 ± 0.51	1.21 ± 0.60	**<0.01***	0.06	0.24	0.26	**0.04***	−0.33	**<0.01***	−0.57
mOFC_R	1.11 ± 0.62	0.97 ± 0.54	1.24 ± 0.66	**0.04***	0.03	0.26	0.24	0.29	−0.20	**<0.01***	−0.45
NAc_R	0.43 ± 0.19	0.35 ± 0.14	0.39 ± 0.16	**0.05***	0.03	**<0.01***	0.44	0.31	0.20	0.21	−0.26
CA_R	0.45 ± 0.10	0.42 ± 0.08	0.46 ± 0.10	**0.04***	0.03	**0.04***	0.38	0.96	−0.04	**0.01***	−0.42

a = ASD-AVP vs. ASD-nAVP, b = ASD-AVP vs. TD, c = ASD-nAVP vs. TD.
η^2:
Small Effect: 0.01–0.06
Medium Effect: 0.06–0.14
Large Effect: ≥ 0.14
Cohen's d:
Small Effect: 0.2–0.5
Medium Effect: 0.5–0.8
Large Effect: ≥ 0.8
ASD-AVP autism spectrum disorder with atypical visual processing, *ASD-nAVP* autism spectrum disorder without atypical visual processing, *TD* typically developing, *PDFF* proton density fat fraction
Statistically significant *p*-values are in bold.
*one-way ANOVA test, *Pfdr* <0.05 indicates a statistically significant difference according to the multiple comparisons correction using the false discovery rate (FDR).
*Turkey honestly significant difference (HSD) $p < 0.05$ represents statistically significant.

Inter-group differences for lipid and myelin contents

PDFF values are presented in Table 2, Table S3, and Fig. 2A. Lipid content represented by fat fraction is expressed in percentage (%). Myelin content (MyC) values are shown in Tables 3, S4, and Fig. 2B, with measurements in milliliters (mL). Statistical analyses revealed co-occurring abnormalities in lipid metabolism and myelin

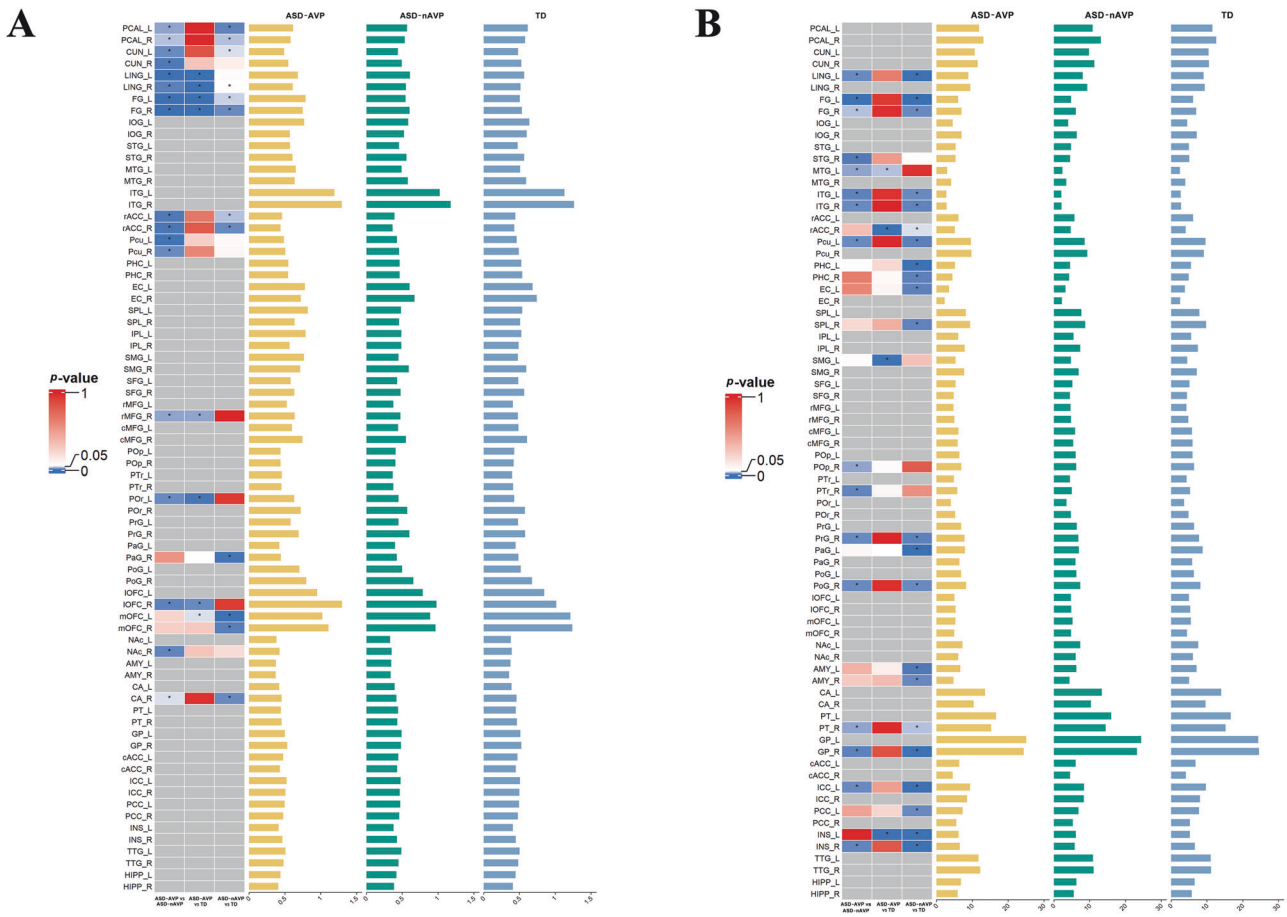

Fig. 2 Regional differences in brain lipid and myelin contents across groups. Regional group differences in proton density fat fraction (PDFF) (**A**) and myelin content (MyC) (**B**). Bar plots show mean PDFF values across brain regions for the ASD-AVP, ASD-nAVP, and TD groups. One-way ANOVA was conducted per region (significance threshold: $p < 0.05$ after FDR correction). Only regions with significant ANOVA results underwent post hoc Tukey's Honestly Significant Difference (HSD) tests; nonsignificant regions are shown in gray. The color scale reflects p-value magnitude. Asterisks indicate significant post hoc intergroup differences ($p^* < 0.05$). ASD-AVP autism spectrum disorder with atypical visual processing, ASD-nAVP autism spectrum disorder without atypical visual processing, TD typically developing.

content within visual processing regions. Specifically, the ASD-AVP group showed elevated lipid levels relative to both the ASD-nAVP and TD groups, while the ASD-nAVP group exhibited reduced lipid content in certain regions, e.g., bilateral lingual gyrus (LING) and rostral anterior cingulate cortex (rACC), compared to TD controls. Regarding myelin content, the ASD-AVP group demonstrated higher MyC than the ASD-nAVP group, while exhibiting broadly lower MyC relative to the TD group.

Correlation analysis of lipid and myelin contents in specific brain regions

Correlations between PDFF and MyC were examined in regions with significant inter-group differences for both measurements (Table 4). A significant positive correlation between lipid and myelin content was observed only in the bilateral FG in the ASD-AVP group (left: $r = 0.47$, $p < 0.01$; right: $r = 0.41$, $p < 0.01$).

Serum biomarker analyses

Serum iron and ceruloplasmin levels were significantly lower in the ASD-AVP group than in both the ASD-nAVP and TD groups, whereas lead levels were higher than in the TD group (see Table 1 for details).

In the ASD-AVP group, serum lead levels correlated positively with lipid content in the bilateral precuneus (Pcu) (left: $r = 0.30$, $p = 0.03$; right: $r = 0.32$, $p = 0.02$) and right CUN ($r = 0.34$, $p = 0.02$). Meanwhile, Fe levels showed negative correlation with lipid content in the bilateral FG (left: $r = -0.40$, $p < 0.01$; right: $r = -0.31$, $p = 0.03$) and broadly negative associations with myelin content across visual and non-visual regions. In addition, Ceruloplasmin levels were negatively correlated with lipid content in the bilateral FG (left: $r = -0.68$, $p < 0.01$; right: $r = -0.47$, $p < 0.01$) and exhibited widely negative correlation with myelin content, largely overlapping with iron–myelin correlated regions (Tables 5, 6, S5, S6, and Fig. 3A). Moreover, a significant positive correlation was observed between serum ceruloplasmin and iron levels exclusively in the ASD-AVP group ($r = 0.33$, $p < 0.01$) (Table S9).

The ASD-nAVP group showed similar but weaker negative correlation trends between blood biomarkers (lead, iron, ceruloplasmin) and lipid/myelin metrics, with more spatially restricted effects (Table 5, Table 7, Table S7, Table S8, and Fig. 3A).

Mediation analyses

In the ASD-AVP group, mediation analysis identified significant pathways through which iron and ceruloplasin influenced myelin content via lipid metabolism in the bilateral FG (Table S10 and Fig. 3B).

Diagnosis power of FG biomarkers

ROC analysis using combined lipid and myelin contents in the bilateral FG as model regressor showed high discriminatory accuracy between ASD-AVP and TD groups (AUC = 0.93, 95% CI

Table 3. MyC differences among the three groups.

	ASD-AVP ($n = 90$)	ASD-nAVP ($n = 89$)	TD ($n = 109$)	one-way ANOVA (*pfdr* /η^2)		*p*-value (Turkey HSD test)/Cohen's d					
						a		b		c	
LING_L	8.86 ± 1.94	8.10 ± 1.51	9.08 ± 1.74	**<0.01***	0.06	**0.01***	0.44	0.64	−0.12	**<0.01***	−0.60
FG_L	6.06 ± 2.40	4.83 ± 1.61	6.15 ± 2.09	**<0.01***	0.08	**<0.01***	0.60	0.94	−0.04	**<0.01***	−0.70
FG_R	6.95 ± 2.47	6.17 ± 1.71	7.02 ± 1.88	**0.03***	0.03	**0.03***	0.37	0.97	−0.03	**0.01***	−0.47
STG _R	5.32 ± 1.72	4.57 ± 1.27	5.08 ± 1.66	**0.02***	0.04	**<0.01***	0.50	0.51	0.15	0.06	−0.34
MTG_L	2.97 ± 1.52	2.42 ± 1.19	2.47 ± 1.30	**0.03***	0.03	**0.02***	0.40	**0.03***	0.35	0.96	−0.04
ITG_L	2.73 ± 1.67	2.11 ± 0.87	2.68 ± 1.46	**0.02***	0.04	**<0.01***	0.47	0.96	0.04	**0.01***	−0.46
ITG_R	2.78 ± 1.7	2.16 ± 0.98	2.80 ± 1.61	**0.02***	0.04	**0.01***	0.45	**<0.01***	−0.01	**0.04***	−0.47
rACC_R	5.09 ± 2.37	4.71 ± 1.61	4.07 ± 1.35	**<0.01***	0.05	0.34	0.19	**<0.01***	0.54	**0.04***	0.43
Pcu_L	9.57 ± 2.41	8.63 ± 2.15	9.61 ± 2.09	**0.02***	0.04	**0.01***	0.41	>0.99	−0.02	**<0.01***	−0.46
PHC_L	5.14 ± 1.92	4.57 ± 1.33	5.55 ± 1.79	**<0.01***	0.05	0.06	0.35	0.22	−0.22	**<0.01***	−0.61
PHC_R	4.48 ± 1.49	4.29 ± 1.27	4.92 ± 1.55	**0.03***	0.03	0.64	0.14	0.08	−0.29	**<0.01***	−0.45
EC_L	3.47 ± 1.61	3.27 ± 1.20	3.88 ± 1.31	**0.03***	0.03	0.60	0.14	0.10	−0.28	**<0.01***	−0.48
SPL_R	9.35 ± 2.33	8.77 ± 2.44	9.76 ± 2.13	**0.03***	0.03	0.21	0.24	0.41	−0.19	**<0.01***	−0.44
SMG_L	5.31 ± 1.65	4.79 ± 1.53	4.47 ± 1.51	**<0.01***	0.05	0.07	0.33	**<0.01***	0.53	0.33	0.21
POp_R	6.91 ± 1.62	6.25 ± 1.56	6.40 ± 1.66	**0.04***	0.03	**0.02***	0.41	0.07	0.31	0.80	−0.09
PTr_R	5.80 ± 1.77	5.06 ± 1.41	5.30 ± 1.73	**0.03***	0.03	**<0.01***	0.46	0.09	0.28	0.55	−0.15
PrG_R	7.80 ± 2.17	6.87 ± 2.10	7.79 ± 2.11	**0.02***	0.04	**0.01***	0.44	>0.99	<0.01	**<0.01***	−0.44
PaG_L	7.90 ± 3.24	7.03 ± 2.40	8.81 ± 2.54	**<0.01***	0.07	0.09	0.31	0.05	−0.32	**<0.01***	−0.72
PoG_R	8.24 ± 1.90	7.37 ± 2.07	8.17 ± 2.01	**0.02***	0.04	**0.01***	0.44	0.97	0.03	**0.01***	−0.40
AMY_L	6.64 ± 1.71	6.31 ± 1.68	7.11 ± 1.75	**0.02***	0.04	0.40	0.20	0.13	−0.27	**<0.01***	−0.47
AMY_R	4.75 ± 1.57	4.41 ± 1.38	5.04 ± 1.54	**0.04***	0.03	0.29	0.23	0.35	−0.19	**<0.01***	−0.43
PT_R	15.24 ± 1.63	14.48 ± 2.28	15.19 ± 1.80	**0.03***	0.03	**0.02***	0.39	0.98	0.03	**0.03***	−0.35
GP_R	24.36 ± 2.10	23.20 ± 3.22	24.56 ± 2.28	**<0.01***	0.05	**<0.01***	0.43	0.85	−0.09	**<0.01***	−0.50
ICC_L	9.38 ± 2.48	8.46 ± 2.19	9.74 ± 1.94	**<0.01***	0.06	**0.01***	0.40	0.49	−0.16	**<0.01***	−0.62
PCC_L	7.28 ± 2.20	6.90 ± 2.52	7.79 ± 1.88	**0.04***	0.03	0.47	0.16	0.23	−0.25	**0.01***	−0.41
INS_L	6.14 ± 1.84	6.16 ± 1.66	5.25 ± 1.36	**<0.01***	0.07	0.99	−0.01	**0.01***	0.56	**<0.01***	0.61
INS_L	6.49 ± 1.50	5.85 ± 1.44	6.61 ± 1.34	**<0.01***	0.05	**<0.01***	0.44	0.82	−0.09	**<0.01***	−0.55

a = ASD-AVP vs. ASD-nAVP, b = ASD-AVP vs. TD, c = ASD-nAVP vs. TD.
MyC = myelin content
η^2:
Small Effec: 0.01–0.06
Medium Effect: 0.06–0.14
Large Effect: ≥ 0.14
Cohen's d:
Small Effect: 0.2–0.5
Medium Effect: 0.5–0.8
Large Effect: ≥ 0.8
ASD-AVP autism spectrum disorder with atypical visual processing, *ASD-nAVP* autism spectrum disorder without atypical visual processing, *TD* typically developing, *PDFF* proton density fat fraction
Statistically significant *p*-values are in bold.
*one-way ANOVA test, *p*fdr <0.05 indicates a statistically significant difference according to the multiple comparisons correction using the false discovery rate (FDR).
*Turkey honestly significant difference (HSD) post hoc test $p < 0.05$ represents statistically significant.

Table 4. PDFF-MyC correlations in group-differentiating regions of ASD-AVP and ASD-nAVP.

	ASD-AVP			ASD-nAVP		
	r	*p*	*pfdr*	r	*p*	*pfdr*
LING_L	0.13	0.24	>0.99	0.25	0.02*	0.04[a]
FG_L	0.47	<0.01*	<0.01[a]	0.09	0.40	0.40
FG_R	0.41	<0.01*	<0.01[a]	0.27	0.01*	0.04[a]
rACC_R	0.03	0.77	>0.99	0.10	0.37	0.40
Pcu_L	0.24	0.02	0.49	0.21	0.05	0.08

Only coefficients (r) with $|r| \geq 0.3$ and FDR-corrected $p < 0.05$ were considered biologically significant.
ASD-AVP autism spectrum disorder with atypical visual processing, *ASD-nAVP* autism spectrum disorder without atypical visual processing, *TD* typically developing, *PDFF* proton density fat fraction, *MyC* myelin content.
[a]*p*fdr < 0.05 indicates a statistically significant difference according to the multiple comparisons correction using the false discovery rate (FDR).
*$p < 0.05$ represents statistically significant results before FDR correction.

Table 5. Significant correlations of Fe, Ceruloplasmin, and Pb with regional PDFF in the ASD-AVP group and ASD-nAVP group.

Group	Blood Biomarker	Brain Region	r	*p*	*pfdr*
ASD-AVP (n = 90)	Iron (Fe)	FG_L	−0.40	<0.01*	<0.01[a]
		FG_R	−0.31	<0.01*	0.03[a]
	Ceruloplasmin	FG_L	−0.68	<0.01*	<0.01[a]
		FG_R	−0.47	<0.01*	<0.01[a]
	Lead (Pb)	CUN_R	0.34	<0.01*	0.02[a]
		Pcu_L	0.30	<0.01*	0.03[a]
		Pcu_R	0.32	<0.01*	0.02[a]
ASD-nAVP (n = 89)	Ceruloplasmin	FG_L	−0.34	<0.01*	0.03[a]
		FG_R	−0.38	<0.01*	0.01[a]

Only coefficients (r) with $|r| \geq 0.3$ and FDR-corrected $p < 0.05$ were considered biologically significant and are presented in this table.
ASD-AVP autism spectrum disorder with atypical visual processing, *ASD-nAVP* autism spectrum disorder without atypical visual processing, *TD* typically developing, *PDFF* proton density fat fraction.
[a]*p*fdr < 0.05 indicates a statistically significant difference according to the multiple comparisons correction using the false discovery rate (FDR).
*$p < 0.05$ represents statistically significant results before FDR correction.

Table 6. Significant correlations of Fe, Ceruloplasmin, and Pb with regional MyC in the ASD-AVP group.

	Brain Region	r	*p*	*pfdr*
Iron (Fe)	LING_L	−0.62	<0.01*	<0.01[a]
	FG_L	−0.85	<0.01*	<0.01[a]
	FG_R	−0.73	<0.01*	<0.01[a]
	STG_R	−0.55	<0.01*	<0.01[a]
	MTG_L	−0.74	<0.01*	<0.01[a]
	ITG_L	−0.36	<0.01*	<0.01[a]
	ITG_R	−0.72	<0.01*	<0.01[a]
	SPL_R	−0.40	<0.01*	<0.01[a]
	SMG_L	−0.49	<0.01*	<0.01[a]
	rACC_R	−0.37	<0.01*	<0.01[a]
	Pcu_L	−0.49	<0.01*	<0.01[a]
	PHC_L	−0.52	<0.01*	<0.01[a]
	PHC_R	−0.36	<0.01*	<0.01[a]
	EC_L	−0.36	<0.01*	<0.01[a]
	Pop_R	−0.45	<0.01*	<0.01[a]
	PTr_R	−0.44	<0.01*	<0.01[a]
	PrG_R	−0.43	<0.01*	<0.01[a]
	PoG_R	−0.39	<0.01*	<0.01[a]
	AMY_L	−0.40	<0.01*	<0.01[a]
	AMY_R	−0.33	<0.01*	<0.01[a]
	PT_R	−0.45	<0.01*	<0.01[a]
	ICC_L	−0.31	<0.01*	0.01[a]
	INS_R	−0.35	<0.01*	<0.01[a]
Ceruloplasmin	FG_L	−0.42	<0.01*	<0.01[a]
	FG_R	−0.40	<0.01*	<0.01[a]
	STG_R	−0.42	<0.01*	<0.01[a]
	MTG_L	−0.44	<0.01*	<0.01[a]
	ITG_L	−0.36	<0.01*	<0.01[a]
	ITG_R	−0.43	<0.01*	<0.01[a]
	SPL_R	−0.35	<0.01*	<0.01[a]
	SMG_L	−0.32	<0.01*	<0.01[a]
	Pcu_L	−0.32	<0.01*	0.01[a]
	PHC_L	−0.31	<0.01*	0.01[a]
	Pop_R	−0.37	<0.01*	<0.01[a]
	PTr_R	−0.37	<0.01*	<0.01[a]
	PrG_R	−0.38	<0.01*	<0.01[a]
	PoG_R	−0.39	<0.01*	<0.01[a]

Only coefficients (r) with $|r| \geq 0.3$ and FDR-corrected $p < 0.05$ were considered biologically significant and are presented in this table.
ASD-AVP autism spectrum disorder with atypical visual processing, *ASD-nAVP* autism spectrum disorder without atypical visual processing, *TD* typically developing, *MyC* myelin content.
[a]*p*fdr < 0.05 indicates a statistically significant difference according to the multiple comparisons correction using the false discovery rate (FDR).
*$p < 0.05$ represents statistically significant results before FDR correction.

[0.90–0.97]; Fig. 4), and between ASD-AVP and ASD-nAVP groups (AUC = 0.87, 95% CI [0.82–0.92]). Discrimination between ASD-nAVP and TD controls was moderate (AUC = 0.80, 95% CI [0.74–0.86]), consistent with shared ASD pathology. Detailed ROC results are provided in Table S11.

Histological validation in BTBR mice

Our analysis revealed a clear spectrum of myelination patterns (Fig. 5): C57 control mice exhibited the expected continuous and intensely stained blue myelin sheaths, indicative of normal structure. In contrast, BTBR-AVP-like mice showed increased myelin density but with a disorganized, fragmented, and non-continuous architecture. Conversely, BTBR-nAVP-like mice displayed pronounced hypomyelination, characterized by sparse and faint blue staining. These results directly recapitulate the distinct myelin alterations observed in the ASD-AVP, ASD-nAVP, and TD groups.

Longitudinal changes in a subset of ASD-AVP patients

Among the seven ASD-AVP children who completed the 2-year follow-up, five showed a clear increase in serum ceruloplasmin (mean $\Delta = +0.073 \pm 0.007$ g/L), accompanied by improvement in SP2 visual scores (mean $\Delta = -5.0 \pm 1.0$). In these responders, PDFF values in key regions with baseline abnormalities, particularly the bilateral FG, demonstrated a marked decrease (mean Δ in left FG = −0.309 ± 0.275; right FG = −0.198 ± 0.111), while MyC remained stable. In contrast, the two non-responder children, who exhibited minimal change in ceruloplasmin and SP2 scores, showed a progressive increase in regional PDFF (mean Δ in left FG = 0.008 ± 0.009; right FG = 0.009 ± 0.008) with a slight increase in MyC. This pattern suggests that amelioration of ceruloplasmin deficiency is associated with a reversal of lipid accumulation and stabilization of myelin content, correlating with improved clinical symptoms.

DISCUSSION

This study identified a distinct lipid–myelin axis pathology in ASD-AVP, revealed through multimodal neuroimaging and serum biomarker profiling. The most striking feature was concurrent lipid accumulation and myelin hyperplasia in key visual regions, particularly the FG, a pattern absent in ASD-nAVP and in TD children. These findings provide the first in vivo evidence linking lipid–myelin dysregulation to a core sensory–perceptual phenotype in ASD, positioning the FG as a critical hub of vulnerability.

The co-occurrence of lipid accumulation and myelin increase in FG and LING suggests a maladaptive response to neuroinflammation and oxidative stress. In this model, microglia-driven lipid peroxidation generates lipid droplets that disrupt neural function [32, 33], while oligodendrocyte precursor cells attempt remyelination in response to altered visual input.

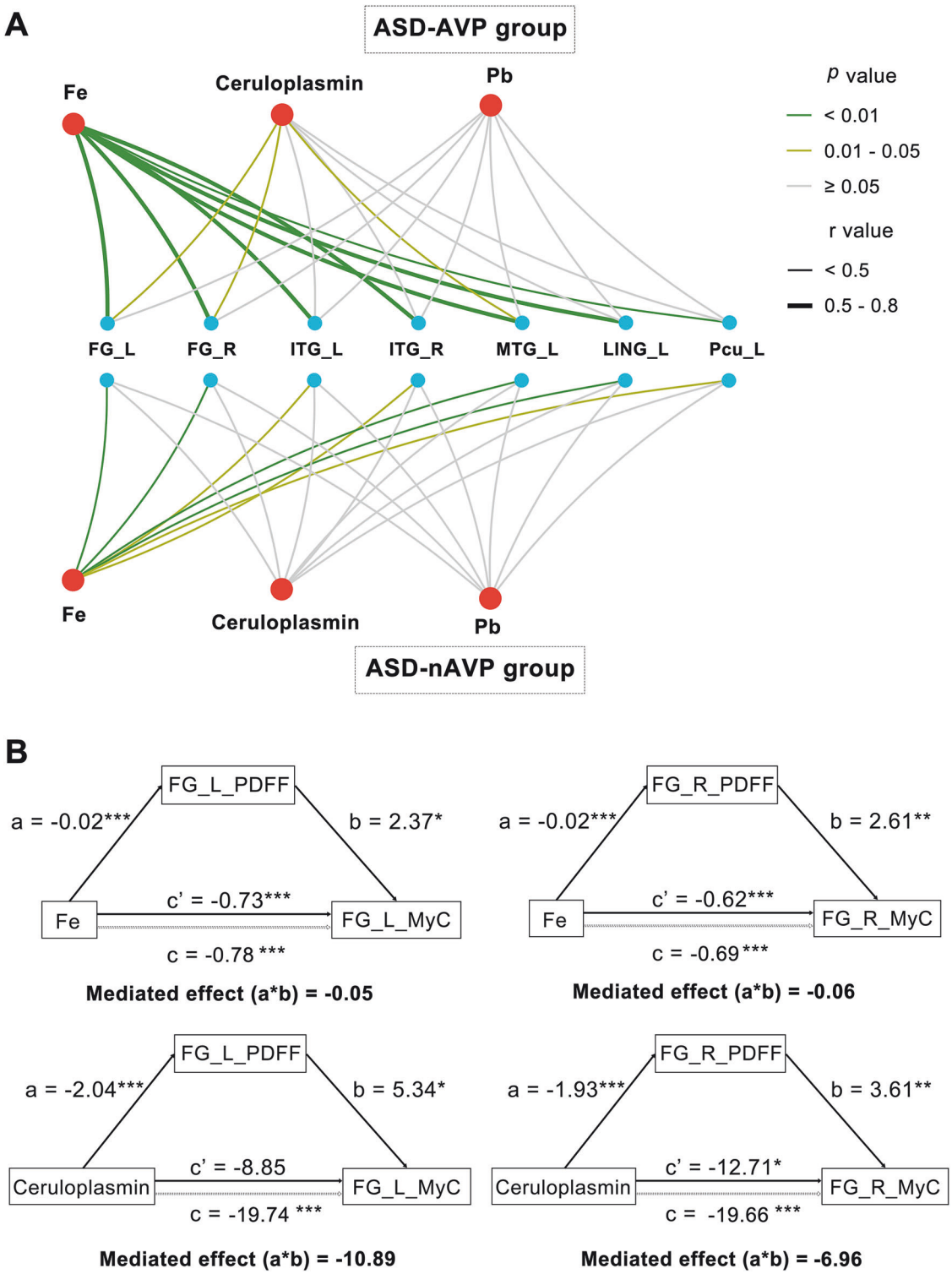

Fig. 3 Associations between blood biomarkers and regional myelin content, and mediation by fusiform lipid content. (**A**) Correlations among serum iron (Fe), ceruloplasmin, lead (Pb), and myelin content (MyC) across brain regions in the ASD-AVP and ASD-nAVP groups. Only the seven regions with the strongest positive correlations (highest r values) are shown for clarity. Dark green: $p < 0.01$; light green: $0.01 \leq p < 0.05$; gray: $p > 0.05$. Lines represent significant associations (FDR-corrected $p < 0.05$), with color indicating biomarker and thickness proportional to correlation strength (r). (**B**) Mediation analysis was performed in the ASD-AVP group to assess the relationship between blood biomarkers (Fe and ceruloplasmin) and MyC in the bilateral fusiform gyrus (FG), using PDFF in bilateral FG as the mediator. Estimates are shown for mediated (a × b), direct (c′), and total (c) effects. *** $p < 0.001$, **$0.001 \leq p \leq 0.01$, *$0.01 < p < 0.05$. FG fusiform gyrus, ITG inferior temporal gyrus, MTG middle temporal gyrus, LING lingual gyrus, Pcu precuneus, PDFF proton density fat fraction, MyC myelin content.

Table 7. Significant correlations of Fe, Ceruloplasmin, and Pb with regional MyC in the ASD-nAVP group.

	Brain Region	r	*p*	*pfdr*
Iron (Fe)	LING_L	−0.41	<0.01*	<0.01[a]
	FG_L	−0.60	<0.01*	<0.01[a]
	FG_R	−0.50	<0.01*	<0.01[a]
	STG_R	−0.32	<0.01*	0.01[a]
	MTG_L	−0.46	<0.01*	<0.01[a]
	ITG_L	−0.39	<0.01*	<0.01[a]
	ITG_R	−0.36	<0.01*	<0.01[a]
	SMG_L	−0.40	<0.01*	<0.01[a]
	Pcu_L	−0.35	<0.01*	<0.01[a]
	Pop_R	−0.33	<0.01*	0.01[a]
	PTr_R	−0.42	<0.01*	<0.01[a]
	PrG_R	−0.35	<0.01*	<0.01[a]
	PoG_R	−0.35	<0.01*	<0.01[a]
	AMY_L	−0.38	<0.01*	<0.01[a]
	AMY_R	−0.30	<0.01*	0.02[a]
	PT_R	−0.40	<0.01*	<0.01[a]
	ICC_L	−0.31	<0.01*	0.01
	PHC_L	−0.41	<0.01*	<0.01[a]
Ceruloplasmin	SMG_L	−0.32	<0.01*	0.01[a]
	PrG_R	−0.25	0.02*	0.05[a]
	PCC_L	−0.31	<0.01*	0.01[a]
Lead (Pb)	PT_R	−0.40	<0.01*	<0.01[a]
	GP_R	−0.37	<0.01*	<0.01[a]

Only coefficients (r) with $|r| \geq 0.3$ and FDR-corrected $p < 0.05$ were considered biologically significant and are presented in this table.
ASD-AVP autism spectrum disorder with atypical visual processing, *ASD-nAVP* autism spectrum disorder without atypical visual processing, *TD* typically developing, *MyC* myelin content.
[a]*pfdr* <0.05 indicates a statistically significant difference according to the multiple comparisons correction using the false discovery rate (FDR).
* $p < 0.05$ represents statistically significant results before FDR correction.

However, the inflammatory milieu likely impairs myelin quality, producing structurally abnormal sheaths [34, 35], a state we describe as inflammatory pseudo-compensation. The LING supports basic visual processing like color and scene recognition [36]; the FG is essential for facial recognition [37]; the Pcu integrates visual and spatial information [38]; and the rACC regulates visual attention and emotional conflict [39]. The functional roles of these regions in face recognition, scene processing, and visual–spatial

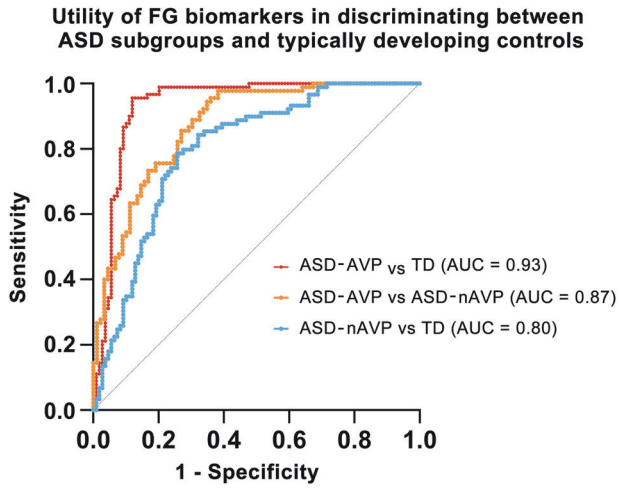

Fig. 4 Diagnostic performance of bilateral fusiform lipid and myelin features. Receiver operating characteristic (ROC) curves demonstrating diagnostic performance of combined lipid and myelin features in bilateral FG for group comparisons: ASD-AVP vs. TD (red; AUC = 0.93, 95% CI [0.90–0.97]); ASD-AVP vs. ASD-nAVP (orange; AUC = 0.87, 95% CI [0.82–0.92]); ASD-nAVP vs. TD (blue; AUC = 0.80, 95% CI [0.74–0.86]). The gray diagonal represents chance discrimination (AUC = 0.5). ASD-AVP autism spectrum disorder with atypical visual processing, ASD-nAVP autism spectrum disorder without atypical visual processing, TD typically developing, FG fusiform gyrus, AUC area under the curve. Combined features were derived using logistic regression.

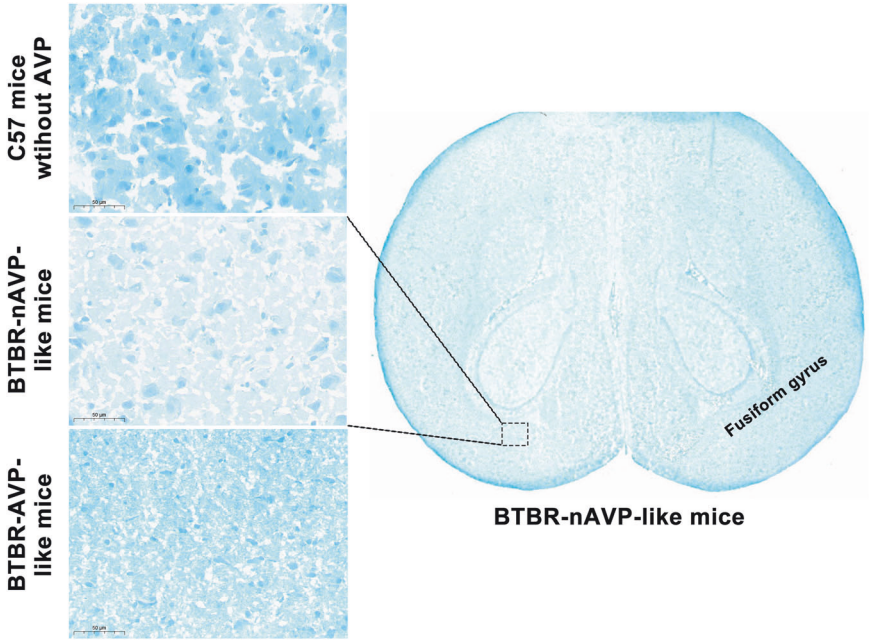

Fig. 5 Luxol Fast Blue (LFB)-stained sections from the fusiform gyrus of C57 (control), BTBR-AVP-like, and BTBR-nAVP-like mice. The C57 control mouse exhibits normal, continuous, and intensely stained blue myelin sheaths. In contrast, the BTBR-AVP-like mouse shows hypermyelination characterized by disorganized and fragmented architecture. The BTBR-nAVP-like mouse displays significant hypomyelination, with sparse and pale blue staining. Scale bar: 50 μm (applicable to all panels).

integration align with the atypical visual behaviors observed in ASD-AVP. Of particular note, the large effect sizes for FG alterations (e.g., d = 2.06 for left FG PDFF) suggest a profound localized metabolic shift, reinforcing its role as a core pathological hub and a promising target for mechanistic intervention. Strikingly, the myelination patterns in the mouse model directly recapitulated the distinct pathology observed in the human subgroups. The disorganized, fragmented, yet hypermyelinated architecture seen in BTBR-AVP-like mice offers a pathological substrate for our concept of "inflammatory pseudo-compensation.

Other visual and socio-cognitive regions showed either myelin increases without lipid change, or lipid accumulation without myelin gain. These patterns may reflect region-specific balances between inflammatory injury, metabolic capacity, and repair potential. For example, early visual cortex regions with lipid accumulation but no myelin gain may represent a "failed repair" stage, where oxidative damage overwhelms remyelination capacity.

A notable feature was the FG-specific positive correlation between lipid and myelin content in ASD-AVP, absent in other regions and groups. We interpret this as evidence of a lipid–myelin metabolic coupling unique to the FG: peroxidized lipids may be repurposed as an energy source for remyelination [33], but sustained demand risks mitochondrial dysfunction and further oxidative injury [40]. This selective vulnerability may stem from the FG's high metabolic rate and its central role in socially relevant visual processing [41].

Serum biomarker profiles in ASD-AVP, reduced iron and ceruloplasmin, and elevated lead, point to systemic oxidative stress as a driver of the observed brain changes. The overlap between biomarker associations and FG pathology supports a self-reinforcing cycle: impaired iron metabolism and ceruloplasmin deficiency promote oxidative damage, which disrupts myelin integrity and fuels further lipid peroxidation [19, 21, 22, 42, 43]. Mediation analysis underscored ceruloplasmin's pivotal role, suggesting that restoring its function could be a therapeutic priority.

The ASD-nAVP/BTBR-nAVP-like subgroup exhibits a consistent cross-species profile of a hypometabolic, hyporepair state. In humans, this is evidenced by widespread reductions in lipid and myelin content alongside weak biomarker-brain associations. This finding is directly recapitulated at the histological level in BTBR-nAVP-like mice, which show pronounced hypomyelination. This profile likely reflects a protective downregulation of metabolic and reparative processes, a "metabolic quiescence" pathway, that limits lipotoxic stress but simultaneously constrains adaptive plasticity [44, 45], potentially explaining the absence of pronounced atypical visual processing in this subtype.

Although changes were also observed in non-visual networks, they were less consistent than those in the visual pathway, reinforcing the relevance of sensory-specific mechanisms in ASD subtyping. The FG emerged as both a mechanistic and diagnostic focal point, with combined lipid–myelin measures showing strong potential for distinguishing ASD-AVP from other groups.

These findings suggest a sequential therapeutic strategy for ASD-AVP: first targeting oxidative stress and lipid peroxidation (e.g., via ceruloplasmin restoration), then correcting iron deficiency. Imaging and biochemical markers from the FG could serve as sensitive tools for monitoring treatment response.

Furthermore, our preliminary longitudinal data in a subset of ASD-AVP patients provide critical within-subject evidence for the proposed mechanism. The observed dissociation—where increasing ceruloplasmin levels were associated with decreased FG lipid content, stabilized myelination, and improved visual symptoms—directly links the modulation of this peripheral metabolic factor to changes in both brain pathology and clinical presentation. Although based on a small sample, these findings strongly suggest that ceruloplasmin is not merely a static biomarker but a dynamic player in the lipid-myelin pathology of the visual ASD subtype, underscoring its potential as a therapeutic target.

Several limitations merit acknowledgment. First, we did not directly measure central iron or ceruloplasmin levels. Additionally, while the BTBR model recapitulates core ASD behaviors and our stratification addresses some heterogeneity, its translatability to human ASD subtypes requires further clinical validation. Also, the absence of left-handed participants may limit the generalizability of our findings to left-handed individuals with ASD. Finally, the cross-sectional design prevents analysis of temporal progression; longitudinal and interventional studies are essential to establish causality and evaluate targeting the lipid–myelin axis for therapeutic benefit.

This study establishes distinct lipid-myelin pathologies underlying ASD visual subtypes: a hypermetabolic, inflammation-driven "inflammatory pseudo-compensation" in ASD-AVP, characterized by FG-centered lipid accumulation and maladaptive myelination, versus a protective metabolic quiescence with hypomyelination in ASD-nAVP. Histopathological validation in BTBR mice confirmed these as divergent mechanisms, disorganized hypermyelination versus profound hypomyelination, rather than a continuous spectrum. The elucidated ceruloplasmin–iron–lipid axis and the FG biomarker signature provide a mechanistic framework for subtyping and future targeted interventions, moving beyond behavioral phenotyping towards biology-driven stratification.

DATA AVAILABILITY

The datasets generated and/or analysed during the current study are available from the corresponding author on reasonable request.

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AUTHOR CONTRIBUTIONS
YY Deng, SS Zhong, SH Wang, JL Zhang, ZP Lin, WY Zhou, HZ Deng, and RM Guo: Data curation, Investigation, Software. X Zhou, BY Yin, FY Zou, JY Zhao, YX Ni, XW Luo, and LS Shen: Data curation, Formal analysis, Methodology. YY Deng, SS Zhong, SH Wang, and RM Guo: Writing - original draft. RM Guo:Funding acquisition, Supervision. Ya-Yin Deng, Shuang-Shuang Zhong, Shi-Huan Wang, and Bo-Ya Yin contributed equally to this work; all of them have made significant contributions in areas such as case collection, data analysis, statistical analysis, and article writing. Corresponding author Ruo-Mi Guo controlled the study data. The authors (JL Zhang, ZP Lin) from GE Healthcare participated in sections of image post-processing and statistical consultation of the original draft.

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COMPETING INTERESTS
The authors declare no competing interests.

ETHICS APPROVAL AND CONSENT TO PARTICIPATE
This study was approved by the Institutional Research Ethics Committee of the Third Affiliated Hospital of Sun Yat-Sen University ([2022]02-005-01). Written informed consent was obtained from the parents or legal guardians of all participants. All methods involving human participants were performed in accordance with the relevant guidelines and regulations and with the ethical principles of the Declaration of Helsinki. All animal procedures were approved by the Institutional Animal Care and Use Committee (IACUC) of The Third Affiliated Hospital of Sun Yat-Sen University (Approval No. IACUC-2023-A014) and were conducted in accordance with the relevant institutional guidelines and regulations and the National Institutes of Health Guide for the Care and Use of Laboratory Animals. Every effort was made to minimize animal suffering and reduce the number of animals used.

CONSENT FOR PUBLICATION
Not applicable.

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Correspondence and requests for materials should be addressed to Wen-Ying Zhou, Hong-Zhu Deng or Ruo-Mi Guo.

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